

Endocrine Physiology, Pathophysiology and Pharmacology: Gonads

Integrated Biomedical and Clinical Sciences: OMFS513

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Mission: To advance the science and art of health through education, service, scholarship and social accountability

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Learning Objectives & Competency Statements

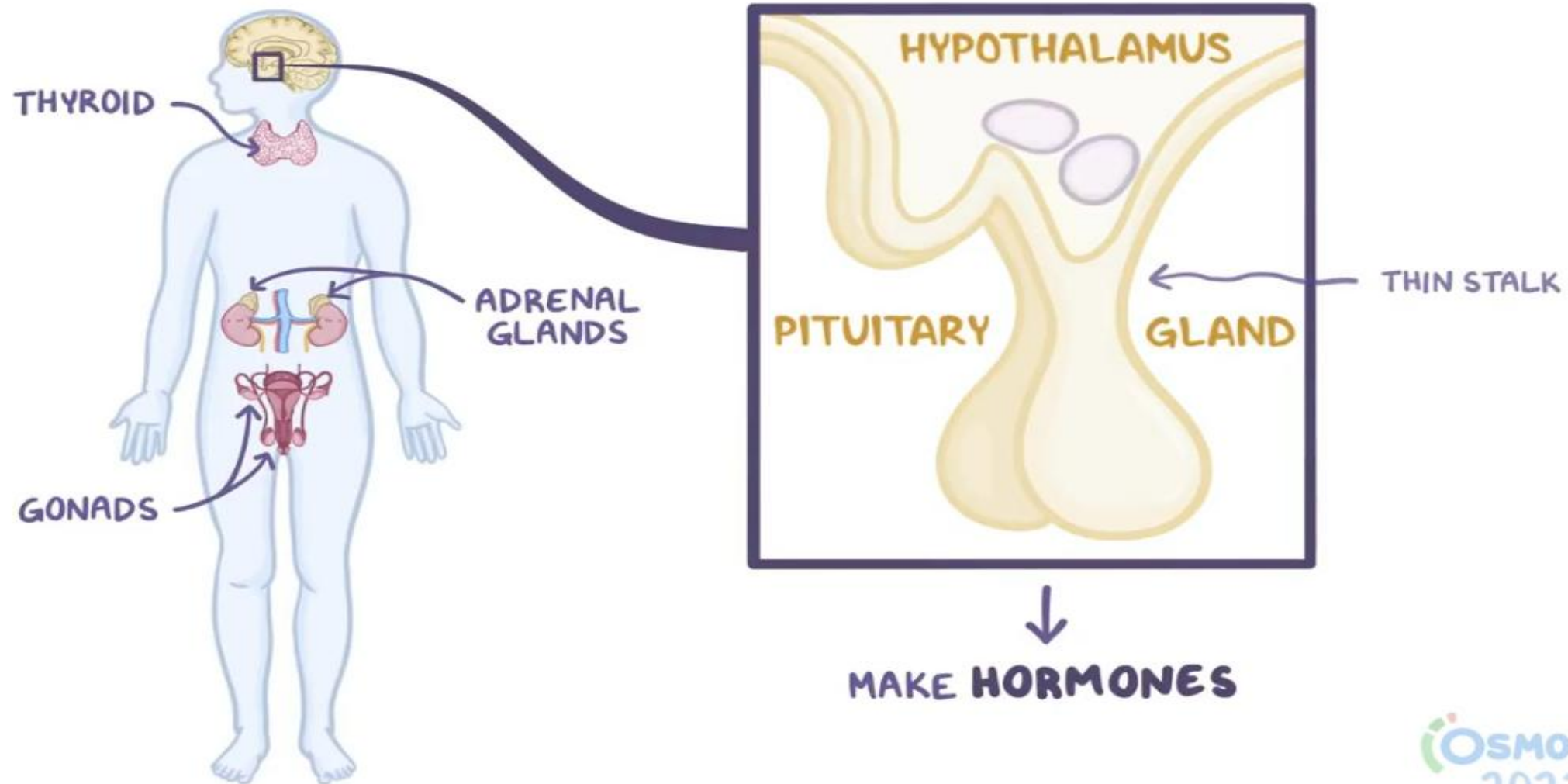
Learning Outcomes

Understand pharmacological basis for drugs affecting the endocrine system with emphasis on gonads

Competency statements

Develop foundational knowledge for underlying health conditions in patients seeking dental care

Endocrine System



Drugs Affecting Thyroid

DRUGS AFFECTING THE THYROID

Iodine and potassium iodide LUGOL'S
SOLUTION

Levothyroxine SYNTHROID

Liothyronine CYTOMEL

Liotrix THYROLAR

Methimazole TAPAZOLE

Propylthiouracil (PTU) GENERIC ONLY

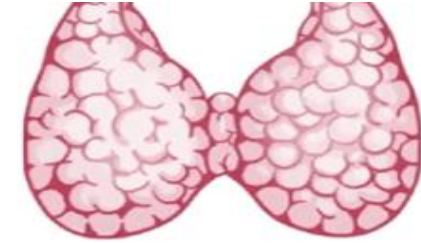
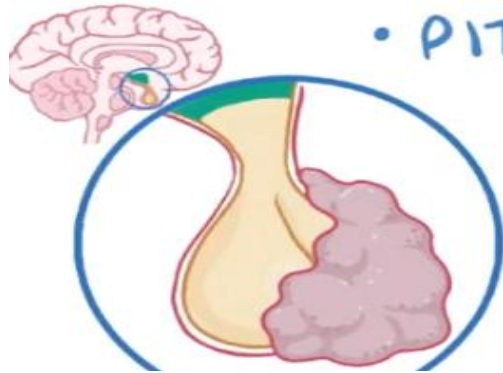
HYPERTHYROIDISM

~ EXCESS THYROID HORMONE

- OVER-FUNCTIONING THYROID GLAND

- PITUITARY TUMOR

- INFLAMMATION of THYROID



SYMPTOMS

~ WEIGHT LOSS (DESPITE INCREASED APPETITE)

↳ ↑ BASAL METABOLIC RATE

~ HEAT INTOLERANCE

↳ BODY PRODUCING MORE HEAT



~ RAPID HEART RATE, SWEATING, HYPERACTIVITY, ANXIETY, & INSOMNIA

↳ EFFECT of THYROID HORMONES ON SYMPATHETIC NERVOUS SYSTEM

Hyperthyroidism Management

A) Targeting thyroid gland by:

- 1- Surgical removal or gland destruction (radioactive iodine)
- 2- Inhibition of thyroid hormone synthesis
- 3- Preventing release of thyroid hormone

B) Treat peripheral symptoms of thyroid hormone (decrease effectiveness)

Example: β -blockers for cardiovascular effects

Thyroid Synthesis Inhibitor

Examples: Thioamides, Oral medications

MOA: concentrate in thyroid and block synthesis of thyroid hormones

Indication: hyperthyroidism treatment

Dental-adverse effects: Altered taste sensation

Methimazole: salivary gland swelling

Propylthiouracil: increases hepatotoxicity risk and hepatic adverse effects

Hormone Release Blocker

Examples: solution of potassium iodide (10%) and iodine (5%) (Lugol's Solution®)

MOA: inhibits iodination of tyrosine during T3 and T4 synthesis, inhibits release of thyroid hormone from thyroglobulin.

Indications: hyperthyroidism, thyroid storm treatment, topical antiseptic solution

Adverse effects: metallic taste

Severe adverse effects: sore mouth and throat, tongue swelling, mucus membrane ulceration

HYPOTHYROIDISM

~ LACK of THYROID HORMONES

CAUSES

• **PRIMARY** ~ THYROID GLAND
↳ NOT MAKING
ENOUGH THYROID
HORMONE

• **SECONDARY**

"central hypothyroidism".
Occur due to tumor on the
anterior pituitary or damage to
the hypothalamus

SYMPTOMS

* CAN TAKE YEARS TO
RECOGNIZE

~ WEIGHT GAIN (DESPITE LOSS of
APPETITE)
↳ ↓ BASAL METABOLIC RATE

~ COLD SENSITIVITY
↳ BODY IS PRODUCING LESS HEAT

~ SLOWER HEART RATE
~ MENTAL SLOWNESS
~ LETHARGY
~ CONSTIPATION

} DECREASED EFFECT of
THYROID HORMONES
ON SYMPATHETIC
NERVOUS SYSTEM

SMOS
2022

Hypothyroidism Management

HYPOTHYROIDISM → ↑↑↑ **T_3 and T_4 LEVELS**

* **IODINE DEFICIENCY**

~FOOD RICH in IODINE



* **PRIMARY, SECONDARY, TERTIARY, or CONGENITAL**

~SYNTHETIC THYROID HORMONE REPLACEMENTS

~ Non-synthetic thyroid hormone derived from porcine source

Synthetic Analogues

-Examples: Levothyroxine (synthroid®), Liothyronin (Cytomel®), liotrix (thyrolar®).

Levothyroxine: synthetic analogue of T4

Liothyronine: synthetic analogue of T3

Liotrix: a mixture of T4 and T 3 in 4:1 ratio

-Indications: hypothyroidism treatment, thyroid supplement following surgical thyroid gland removal

Liotrix: used to treat/prevent enlargement of thyroid gland

Levothyroxine: medication of choice for long term hypothyroidism treatment

Synthetic Analogues

- Adverse effects:** Cardiovascular adverse events (e.g. tachycardia), heat intolerance, anxiety. Liothyronine: **contraindicated** in patients with prior history of cardiovascular diseases
- Liothyronine: more potent, faster onset of action** (enters cells and do the T3 action directly), **shorter duration**
- Levothyroxine: longer duration of action, less potent.** Needs to be converted to T3 by cells. **Better tolerated** by patients with **less side effects** (heat intolerance, anxiety)

Drugs Affecting Adrenal Gland

CORTICOSTEROIDS

Betamethasone CELESTONE, DIPROLENE

Cortisone GENERIC ONLY

Dexamethasone DECADRON

Fludrocortisone GENERIC ONLY

Hydrocortisone CORTEF

Methylprednisolone MEDROL

Prednisolone ORAPRED, PEDIAPRED

Prednisone DELTASONE

Triamcinolone KENALOG, NASACORT,
ARISTOSPAN

INHIBITORS OF ADRENOCORTICOID BIOSYNTHESIS OR FUNCTION

Eplerenone INSPRA

Ketoconazole NIZORAL

Spironolactone ALDACTONE

Glucocorticoids

Suffix: -asone, olone

Examples: cortisone, betamethasone, hydrocortisone

-Indications:

- 1- Replacement hormone therapy for adrenal insufficiency
- 2- Suppress inflammation
- 3- Autoimmune disease
- 4- Treatment of allergies
- 5- Acceleration of lung maturation in pre-mature fetus

Glucocorticoids

- Administered via topical or systemic route
- Classified according to duration of action: short, intermediate, long-acting corticosteroids
- Adverse effects:** correspond to drug dose and interval of treatment
- Side effects:** increased risk of infection, peripheral edema, increased risk of diabetes/hypertension/osteoporosis
- Patients advised against sudden discontinuation due to serious side effects: suppression of hypothalamus, pituitary, adrenal axis (can lead to potentially fatal adrenal insufficiency)

Adrenal Hormone Synthesis Inhibitors

Examples: Etomidate (Amidate),
Ketoconazole (Nizoral), spironolactone
(Aldactone)

MOA:

-Ketoconazole “antifungal”, inhibits adrenal steroid synthesis, used for hyperadrenalism (e.g. Cushing syndrome)

-Etomidate “anesthetic drug”, inhibits steroid synthesis, via IV route for emergency treatment of Cushing syndrome

CORTICOSTEROIDS

Betamethasone CELESTONE, DIPROLENE

Cortisone GENERIC ONLY

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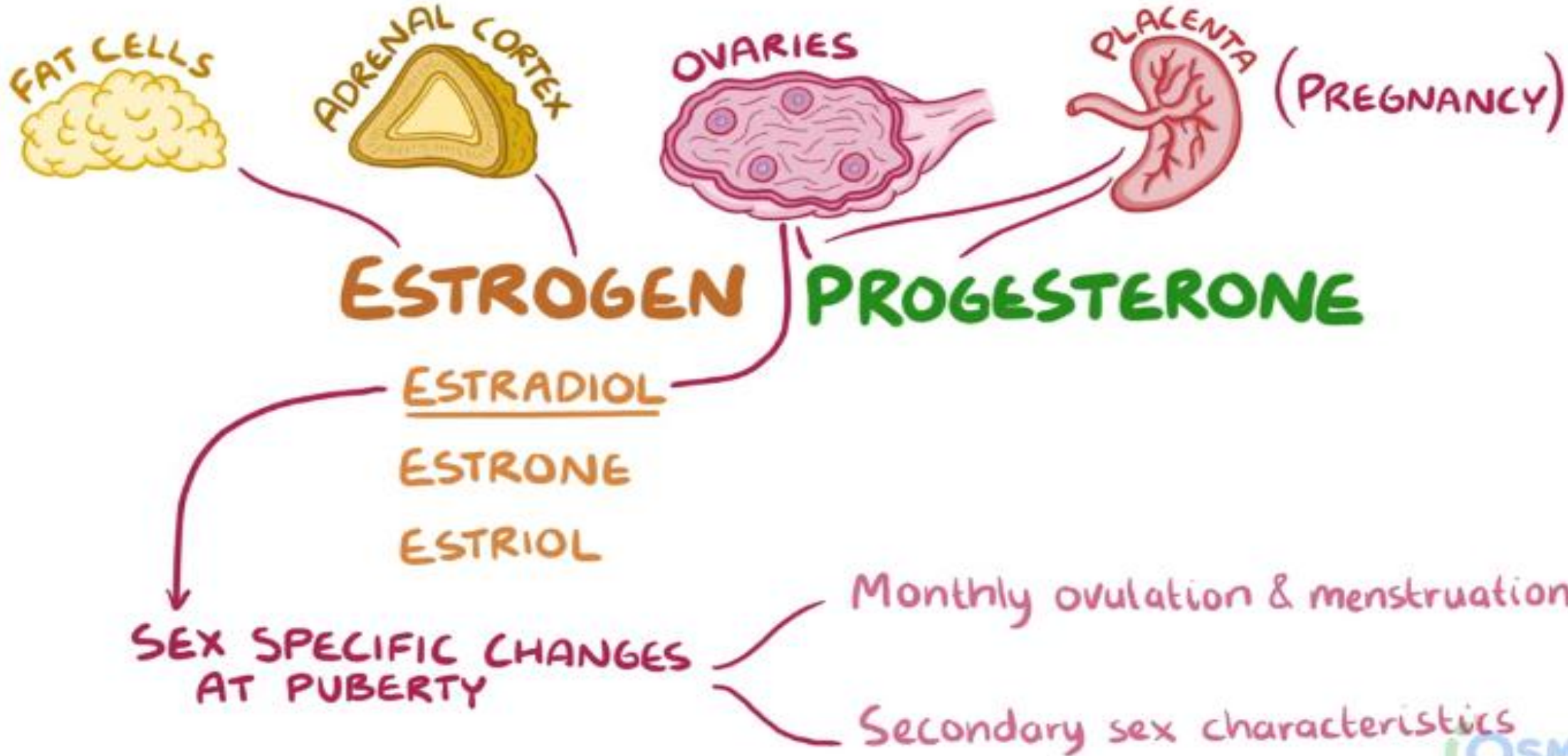
INHIBITORS OF ADRENOCORTICOID BIOSYNTHESIS OR FUNCTION

Eplerenone INSPRA

Ketoconazole NIZORAL

Spironolactone ALDACTONE

FEMALE SEX HORMONES



MAIN FEMALE SEX HORMONES: ESTROGEN & PROGESTERONE

↳ Crucial role from PUBERTY

REPRODUCTIVE PERIOD

GRANULOSA CELLS



ESTROGEN & PROGESTERONE

PREGNANCY

PLACENTA



PROGESTERONE

ESTRIOL

AFTER MENOPAUSE

FAT CELLS



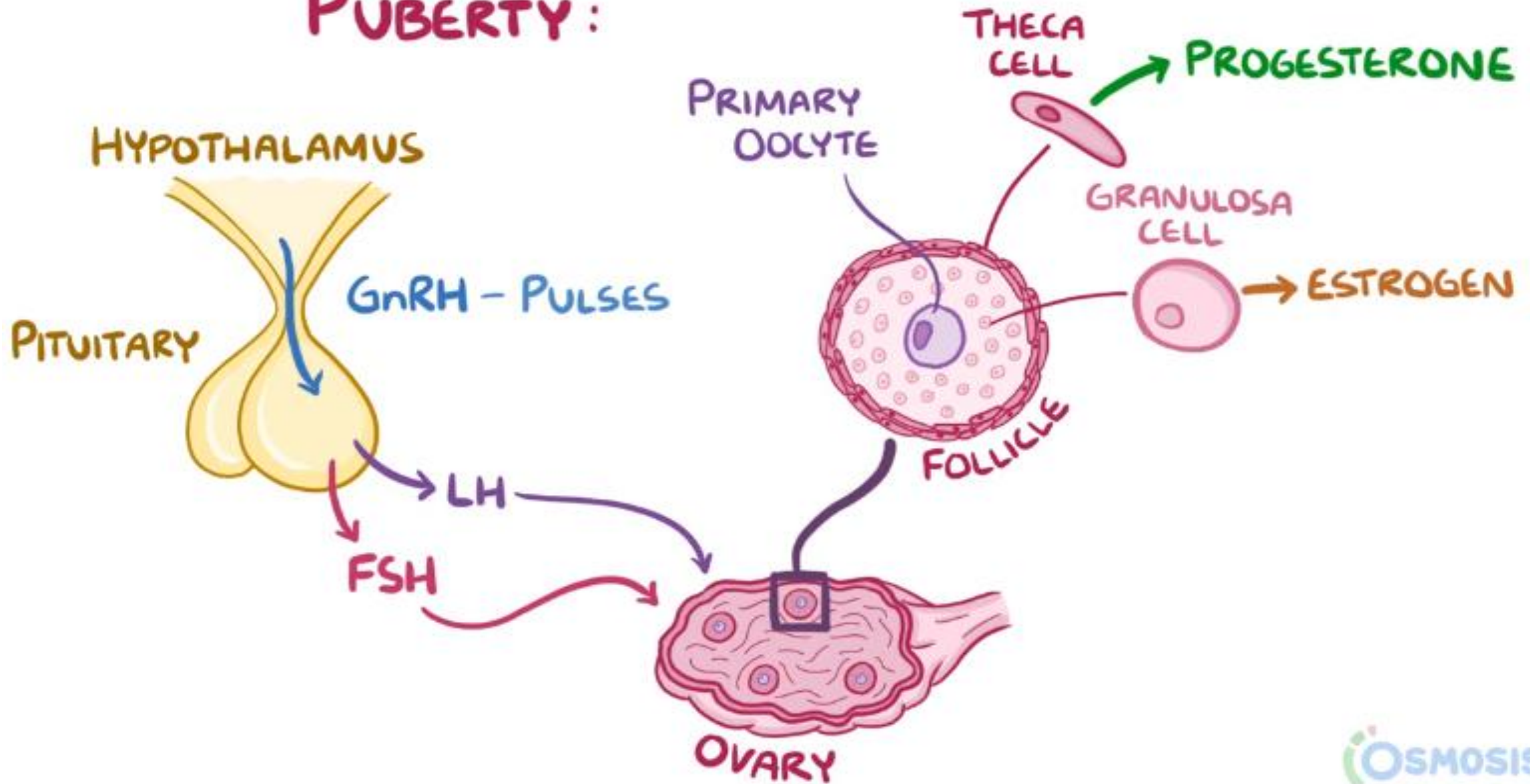
ADRENALS



ESTRONE

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PUBERTY:



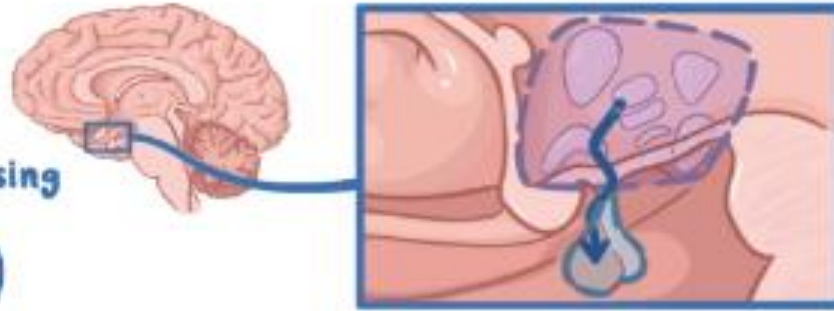
NATURAL ESTROGEN

HYPOTHALAMUS

- Gonadotropin-Releasing Hormone (GnRH)

PITUITARY

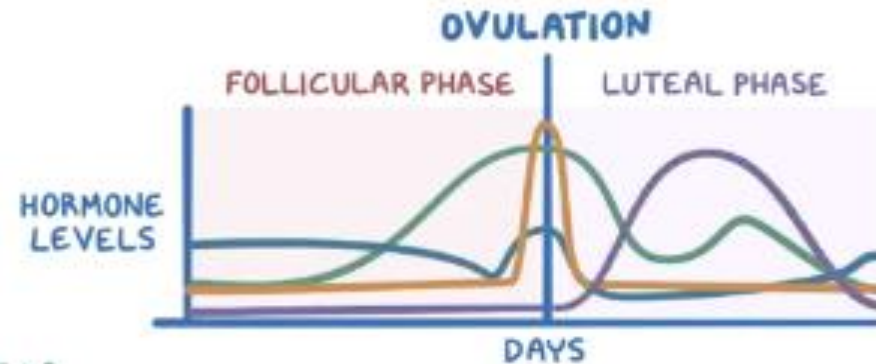
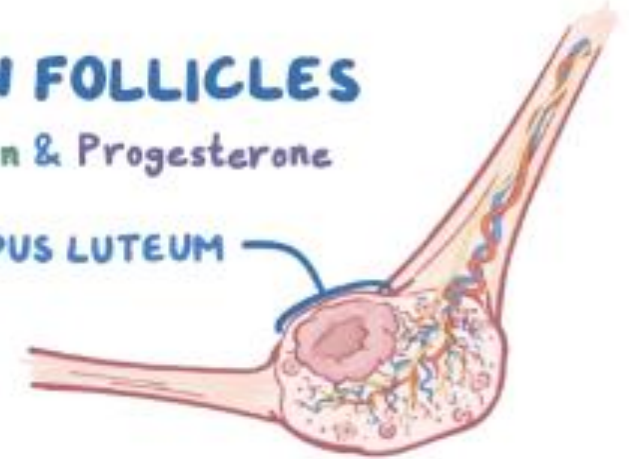
- Follicle-Stimulating Hormone (FSH)
- Luteinizing Hormone (LH)



OVARIAN FOLLICLES

- Estrogen & Progesterone

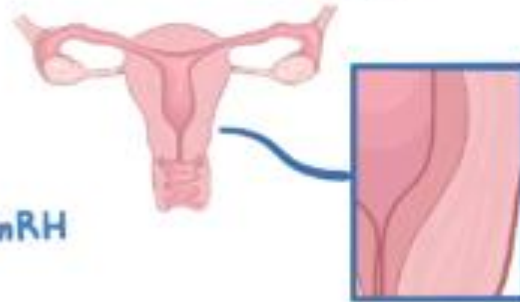
CORPUS LUTEUM



- ESTROGEN
- FSH
- PROGESTERONE
- LH

HIGH LEVELS of ESTROGEN

- THICKENING of the ENDOMETRIUM
- GROWTH of ENDOMETRIAL GLANDS
- EMERGENCE of SPIRAL ARTERIES
- PITUITARY → MORE SENSITIVE to GnRH



FOLLICLE → CORPUS LUTEUM

- SYNTHESIZES PROGESTERONE
- MAINTAIN UTERINE LINING

SYSTEMIC ESTROGEN

* **KEEPS BLOOD VESSEL WALLS FLEXIBLE**

↳ ↓ LDL

* ↓ **OSTEOCLASTS**
* ↑ **OSTEOBLASTS** } SUSTAIN BONE DENSITY



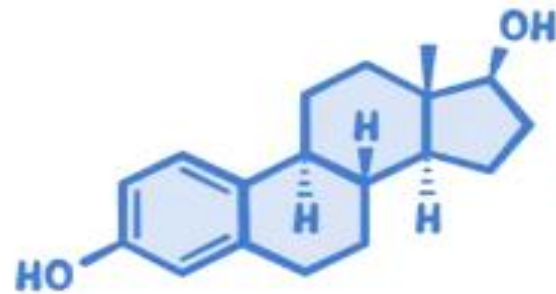
-Estrogen: protective effect on the cardiovascular system, keeps the blood vessels' walls flexible to accommodate blood flow. On the skeleton, helps sustain bone density. Protects against cardiovascular disease, by lowering LDL-Cholesterol.

-Progesterone: plays a role in bone strength and keeping the skin elastic.

Therapeutic Uses

SYNTHETIC ESTROGENS

↳ Act like natural ESTROGENS



**PRIMARY
HYPOGONADISM**



**PRIMARY
AMENORRHEA**



**ESTROGEN
DEFICIENCY**



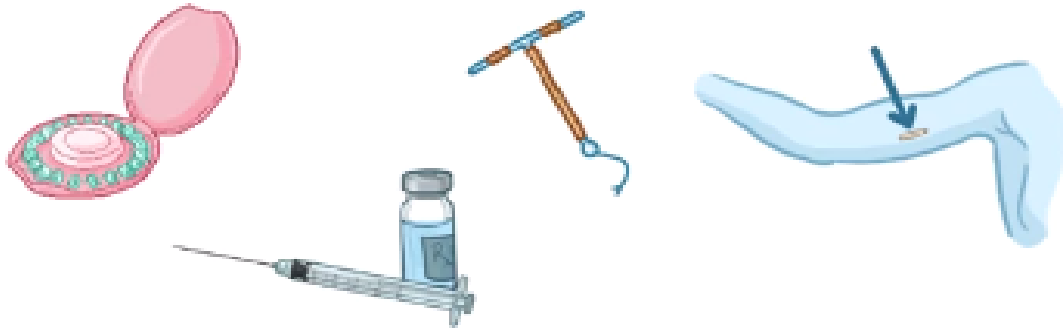
CONTRACEPTIVES



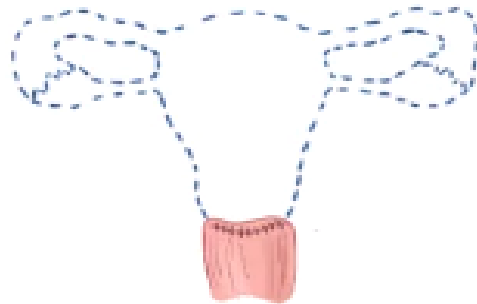
INDICATIONS

CONTRACEPTIVES

↳ ALONE or COMBINATION w/ ESTROGEN



HYSTERECTOMY → NO PROGESTIN



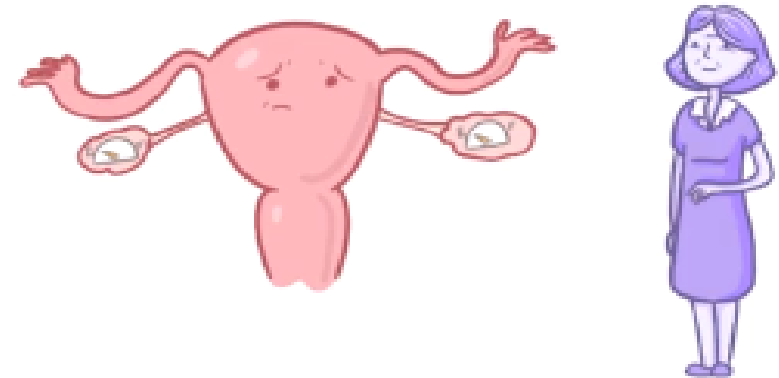
HORMONE REPLACEMENT THERAPY

- WOMEN w/ an INTACT UTERUS

↳ COMBINATION w/ ESTROGEN



PREVENT ENDOMETRIAL HYPERPLASIA



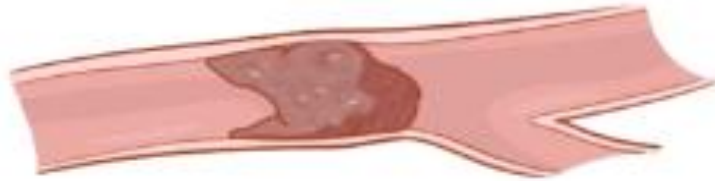
*** ALL WOMEN w/ an INTACT UTERUS
NEED PROGESTERONE + ESTROGEN ***

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Adverse Effects

- ~ VAGINAL BLEEDING
- ~ NAUSEA
- ~ BREAST TENDERNESS

↑ RISK of THROMBOEMBOLIC EVENTS



SMALL ↑ RISK of BREAST CANCER



↑ RISK of ENDOMETRIAL CANCER

↳ WHEN GIVEN w/o PROGESTERONE

FEMALES EXPOSED to DES in UTERO

- VAGINAL ADENOSIS
- CLEAR CELL ADENOCARCINOMA

PROGESTINS = SYNTHETIC PROGESTOGENS

└ SIMILAR EFFECTS to NATURAL PROGESTERONE

**PROGESTERONE
DERIVATIVES**

**TESTOSTERONE
DERIVATIVES**

CONTRACEPTIVES



HORMONE REPLACEMENT
THERAPY (HRT)



TREATMENT of VARIOUS
GYNECOLOGIC CONDITIONS



SIDE EFFECTS

* TOXICITY = LOW at TYPICAL DOSES

- FLUID RETENTION
- ↑ BLOOD PRESSURE
- FATIGUE
- MOOD CHANGES & DEPRESSION
- BREAST DISCOMFORT
- CHANGES in LIBIDO
- MENSTRUAL IRREGULARITIES
(BREAKTHROUGH BLEEDING)



* HIGH DOSES for a LONG PERIOD of TIME

- ↓ BONE DENSITY
- DELAY in OVULATION

Mechanism of Action

- Bound to plasma proteins to circulate
- Unbound or free hormone can enter cell by diffusion and binding to receptors
- Activated steroid-receptor complex interacts with nuclear chromatin to initiate hormone-specific RNA synthesis
- Synthesis of specific proteins that mediate physiologic functions

Hormone Antagonists and Partial Agonists

USED to TREAT

SYNTHETIC ESTROGENS ~ ETHINYL ESTRADIOL ~ DIETHYLSTILBESTROL ~ MESTRANOL	- PRIMARY HYPOGONADISM - PRIMARY AMENORRHEA - HORMONE REPLACEMENT THERAPY - CONTRACEPTIVES
SERMs → AGONISTS in SOME TISSUES ANTAGONISTS in OTHERS ~ TAMOXIFEN	- ER-POSITIVE BREAST CANCER
~ RALOXIFENE	- OSTEOPOROSIS
~ CLOMIPHENE	- INFERTILITY
FULL ANTAGONISTS ~ FULVESTRANT	- BREAST CANCER RESISTANT to TAMOXIFEN

Drug Interactions

- **Estrogen:** could increase effects of corticosteroids
- **Progestins:** some hepatic enzyme-inducing medications decrease progestins' effects
- **Oral contraceptives and antibiotics:** antibiotics may reduce oral contraceptive efficacy
- **Androgens:** effects of anticoagulants, antidiabetics, insulin, and cyclosporine are increased with androgens. Using androgens and corticosteroids can increase edema and exacerbate cardiac or hepatic disease.

Pregnancy or Menopause: Dental Implications

First step: thorough history including asking key questions from physician and patient.

***Patient:**

Pregnancy:

- What trimester?
- Any sign or symptoms of a high-risk pregnancy (e.g. hypertension)
- Any intraoral changes?

Menopause:

- Any oral pain/dryness/burning?

***Physician:**

- Delivery date?
- Is it a high-risk pregnancy?
- Any special concerns?
- What medications for pain control, antibiotic, anesthetic?



Fig. 2. Clinical presentation of a PG on the maxillary interdental gingiva. Photograph attribution: Kasama Kanpittaya, ID: 1421389106.

Dental Implications

***Elective dental treatment during pregnancy:**

- Recommended timing: second trimester or early half of the third trimester

***Emergency dental treatment during pregnancy:**

- As needed any time during pregnancy
- May require a consultation with obstetrician
- Concerns about medications required or effect of treatment on fetus

***Dental radiographs during pregnancy:**

- Untreated dental infections may pose a greater risk to fetus than potential radiation exposure
- Take every precaution to minimize any radiation exposure including use of digital radiography and protective shielding

Dental Implications

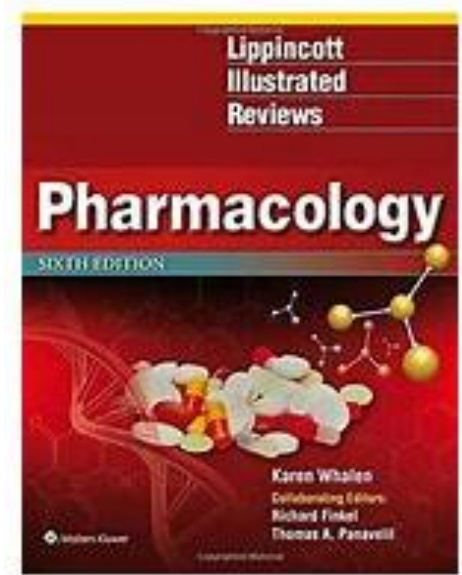
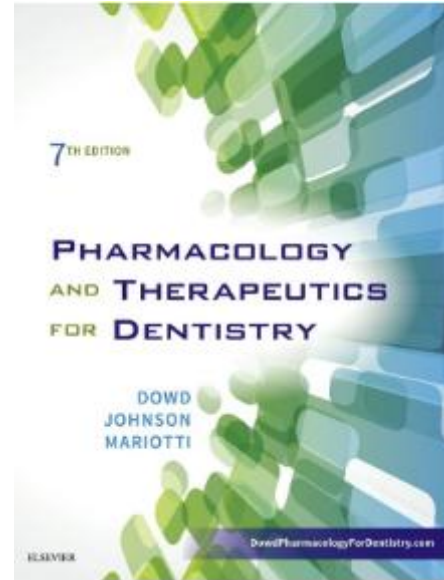
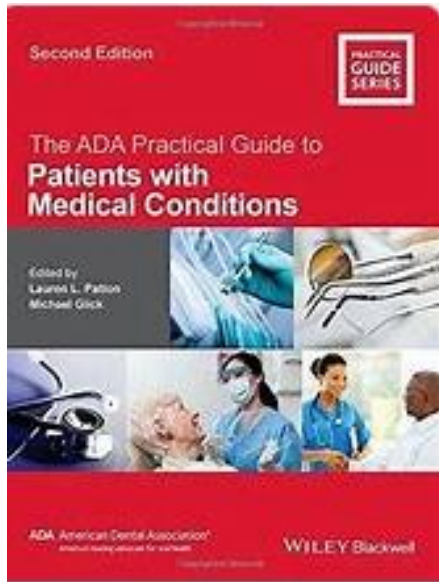
-Early oral hygiene education, Cleaning and improved OH care, More frequent recalls, Eliminating infection, Nutritional counseling, Tobacco cessation, Educate mothers on infants/young children oral health care, Treatment disruptions for urination (fetus pressure on bladder), Difficulty sitting in a reclining or semi-supine position due to compression on superior vena cava (maternal hypotension, decreased cardiac output, loss of consciousness): Turning patient to her left side to relieve pressure. Prescribing antibiotics to a woman taking oral contraceptives: potential interaction: decreased efficacy of oral contraceptives

***Contraception use:**

-Increased body's inflammatory response to oral irritants, Increased risk of dry socket and post-op pain

***Menopausal oral mucosal symptoms:** Pain or burning sensation, altered taste, dryness: palliative treatments

Resources



[Kaleeny, Joseph & Janis, Jeffrey. \(2024\). Pyogenic Granuloma Diagnosis and Management: A Practical Review. Plastic and Reconstructive Surgery - Global Open. 12. e6160.](#)

[10.1097/GOX.00000000000006160.](#)

[Estrogen and progesterone | Osmosis](#)

[Estrogens and antiestrogens | Osmosis](#)

[Progestins and antiprogestins | Osmosis](#)

Applications

Background and/or Patient History

Severe lupus erythematosus.

Medications

30 mg of prednisone/day for autoimmune disease treatment

Current Findings

Rampant decay on all teeth which require full mouth extraction and denture delivery.